any JavaScript program is valid TypeScript, but TypeScript also allows you to add some basic type checking to your code. More advanced usage of TypeScript allows you to use the type system to prevent invalid states at compile time, which can make runtime errors less likely.

- esbuild calls itself “an extremely fast JavaScript bundler.”[4] The purpose of esbuild is to convert developer-friendly front-end code into a browser-friendly package. The inputs are the code you write—which for our purposes are mostly JavaScript and TypeScript—all arranged in a hopefully logical structure. esbuild packages all the files into a bundle that the browser can use, which involves translating code and resolving references to code in different files. The converted JavaScript files can then be sent to a browser. esbuild is very fast and simpler to configure for basic tasks than other tools.

- JavaScript Bundling for Rails[5] is a new Rails 7.0 tool that allows you to use the JavaScript bundling tool of your choice—we’re using esbuild—to wrap your JavaScript code and prepare it for a Rails asset pipeline.

- CSS Bundling for Rails[6] is a new Rails 7.0 tool that allows you to use one of a few different CSS processing tools—we’re using Tailwind CSS—to convert your developer CSS to prepare it for asset download.

- Propshaft[7] is a new Rails 7.0 tool that takes the output of the bundling tools and delivers it to the browser, providing for digest hash creation, a developer platform, and the ability to reference assets in code. It’s a simpler replacement for Sprockets, the older asset pipeline tool, for a toolchain where the asset handler needs to do much less work.

- Import Maps for Rails[8] is a standard tool that allows a browser to import all of your separate JavaScript modules individually, but still allows them to reference each other. This allows JavaScript code to be used without a bundling step. Rails 7 supports Import Maps (in fact, it’s the default), but we won’t use it throughout most of the book because TypeScript and React support is tricky. I’ll discuss this more in Chapter 7, *Bundling*.